

LLM Evaluation Analytics Dashboard

Problem Statement Document-Phase 1, Week 1

Project Area: Artificial Intelligence / Machine Learning / Large Language Models and Data Analytics

Primary Focus: LLM Evaluation, Model Comparison & Performance Analytics

1. Background

Large Language Models (LLMs) such as GPT, Gemini, Claude, and other generative AI models are increasingly being used in applications such as customer support, education, content generation, software development, and information retrieval. As organizations adopt multiple LLMs, evaluating their performance and selecting the most suitable model for a particular task has become an important challenge.

LLM evaluation is difficult because model responses can differ in terms of **accuracy, relevance, coherence, factuality, safety, response time, and cost**. Traditional evaluation methods often depend on manual inspection or simple numerical metrics, which may not provide sufficient insight into why one model performs better or worse than another.

Interactive visual analytics provides an effective approach for understanding large-scale LLM evaluation results. By combining automated evaluation with interactive visualizations, users can compare model responses, identify performance patterns, and explore differences across prompts and evaluation criteria.

2. Problem Context

Organizations and developers commonly test LLMs using manually selected prompts, benchmark datasets, and individual evaluation metrics. These approaches are useful, but they can become difficult to manage when evaluating multiple models across a large number of prompts. Manually comparing thousands of responses is time-consuming and may also introduce inconsistencies in evaluation.

Another challenge is that an LLM may perform well on one type of task while performing poorly on another. A model may provide highly accurate responses but have higher cost or latency, while another model may be faster and cheaper but produce less reliable answers. Therefore, simply assigning an overall score may not provide enough information for making an informed model-selection decision.

Existing evaluation approaches can also make it difficult to understand **when and why one LLM outperforms another**. There is therefore a need for an interactive analytics system that can systematically collect evaluation results, compare multiple LLMs,

visualize their performance, and allow users to explore the results according to different criteria.

3. Core Problem Statement

Existing LLM evaluation approaches can provide useful performance measurements, but their effectiveness may be limited by **manual evaluation, large-scale response analysis, multiple evaluation criteria, model variability, lack of interpretability, and difficulty in comparing different LLMs**.

Therefore, there is a need for an **interactive LLM evaluation analytics system** that systematically evaluates and compares multiple Large Language Models, analyzes their responses using appropriate evaluation metrics, and presents the results through meaningful visualizations.

The proposed system aims to help users understand **which model performs better, under which conditions, and why**, while also considering important factors such as **response quality, accuracy, safety, latency, token usage, and cost**. This can support developers, researchers, and organizations in making more informed decisions about selecting, deploying, and improving LLM-based applications.

4. Research Problem

The research problem is not simply to determine which Large Language Model (LLM) produces the best response. The project must investigate whether different LLMs can be **systematically evaluated and compared across multiple dimensions** while providing meaningful and interpretable insights about their performance.

Challenge	Research concern
Model dependency	Different LLMs may perform differently depending on the task, prompt, and underlying model characteristics.
Evaluation scalability	Manually evaluating a large number of LLM responses is time-consuming and difficult to maintain consistently.
Evaluation metrics	LLM quality cannot be represented by a single metric; accuracy, relevance, coherence, safety, latency, and cost may all need consideration.
Generalization	A model that performs well on one dataset or task may perform differently on unseen prompts or domains.
LLM-as-a-Judge reliability	Automated evaluation using another LLM as a judge may introduce errors, inconsistencies, or evaluator bias.
Interpretability	Users need to understand not only the evaluation score but also why models perform differently.

Performance and cost	A model with high response quality may have greater latency or API cost, creating a practical trade-off.
Data visualization	Large-scale evaluation results can be difficult to interpret without effective interactive visual analytics.

Therefore, there is a need for an evaluation system that can **compare multiple LLMs, measure their performance using appropriate metrics, and provide interactive visual insights into the evaluation results.**

5. Proposed Problem-Solving Direction

The proposed project will develop an **interactive LLM Evaluation Analytics Dashboard** in which user-provided prompts or evaluation datasets are submitted to multiple LLMs. Their generated responses are collected and evaluated using predefined evaluation metrics and, where appropriate, an **LLM-as-a-Judge** approach.

The system will process the evaluation results and present them through interactive charts, graphs, tables, and comparison views. Users will be able to examine individual responses as well as overall model performance and identify patterns across different prompts, tasks, and metrics.

Multiple LLMs can be compared based on factors such as **accuracy, relevance, coherence, factuality, safety, response time, token usage, and estimated cost.**

Proposed high-level workflow:

User/Dataset → Prompt Processing → Multiple LLM APIs → Response Collection → Evaluation/LLM-as-a-Judge → Metric Calculation → Data Processing → Interactive Visualization → Model Comparison → Insights/Reports

6. Project Objectives

1. **Study existing LLM evaluation approaches** through selected research literature.
2. **Identify important evaluation metrics** for measuring LLM response quality and performance.
3. **Prepare and preprocess suitable prompt/evaluation datasets** for testing different LLMs.
4. **Integrate multiple LLMs** and generate responses using standardized prompts.
5. **Implement automated evaluation methods**, including LLM-as-a-Judge where appropriate.

6. **Evaluate and compare LLMs** using metrics such as accuracy, relevance, coherence, factuality, safety, latency, token usage, and cost.
 7. **Develop an interactive web-based dashboard** for exploring and comparing LLM evaluation results.
 8. **Provide visual analytics** using charts, graphs, tables, and filters for easier interpretation of results.
 9. **Investigate model performance across different tasks and datasets** to understand generalization and performance variations.
 10. **Provide interpretable results and reports** that help users identify strengths, weaknesses, patterns, and potential areas for model improvement.
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7. Scope of the Problem

In scope: Prompt/dataset management, LLM API integration, response collection, response evaluation, LLM-as-a-Judge, evaluation metrics, model comparison, performance analysis, cost and latency analysis, interactive visualization, filtering and data slicing, evaluation history, and web-based dashboard/reporting.

Initially out of scope: Training a new Large Language Model from scratch, modifying the internal parameters of commercial LLMs, guaranteed detection of all hallucinations or biases, complete replacement of human evaluation, and direct control over third-party model training or fine-tuning.